



nomen

Credit history that no one has to be trusted about.

Every record on Creditcoin exists because a precompile verified the Ethereum mainnet transaction that produced it. No oracle. No indexer. No self-report.

nomen.ochinimus.app · github.com/seekdaseek/nomen · RWA track

On-chain credit history has one weak link

Whoever writes the record

A lender's repayment history reaches a chain because an operator, an oracle or an indexer put it there. The record is only as good as that party. You cannot check it yourself, and neither can the next lender.

That is the gap Creditcoin exists to close, and until now closing it still required trusting the recorder.

What nomen writes instead

A raw Ethereum transaction and its inclusion proof go to the contract. The native verifier checks it against the attested chain. Only then is the receipt decoded and a record written.

A proof that does not verify writes nothing and costs the sender the gas. There is no owner, no admin, no pause, no edit.

How a record comes to exist

- 1 A borrow, repayment or liquidation happens on Aave v3, Morpho Blue or Spark
- 2 Creditcoin attests the Ethereum block, 23 to 40 blocks behind the head
- 3 The prover returns the encoded transaction, a Merkle inclusion proof and a continuity proof
- 4 Anyone calls `Nomen.record` with those bytes. Anyone. There is no permission
- 5 The verifier precompile checks inclusion and continuity. Nothing from the caller is trusted
- 6 The receipt is decoded, the emitter and layout checked, the borrower's record updated

MEASURED ON CREDITCOIN CC3 TESTNET, EVERY FIGURE FROM A RUN

Running against the live market

369

credit events proven and recorded

\$44.2M

notional across 344 priced events

142

borrowers scored on-chain

0

failed records

353 loans opened · 9 repayments · 7 liquidations Aave v3, Morpho Blue and Spark

A 7,200-block backlog of 2,548 events scanned in 8 seconds. The submitter recorded 75 events in its first six minutes at six per batch, 296k gas average, zero failures. The ledger keeps one row per observed event through seen, attested, proven, recorded or failed with its reason, and rows are never deleted. Every number on the site and in this deck is read from that ledger.

Attestcoin is not a component, it is the premise

```
Nomen.record( kind, chainKey, height, txBytes, merkleRoot, siblings, lowerEndpointDigest, continuityRoots )
```

Same signature as ASCBase.execute. Same internal query id, replay check and proof verification, used verbatim. The verifier precompile at 0xFD2 decides whether the transaction was in an attested Ethereum block.

Gas per record

214k to 414k, growing with receipt size and roughly 1,100 gas per continuity root

Cost per record

0.0001 to 0.0002 CTC. Gas caps nothing; attestation lag and the log source do

Rejected proof

estimateGas runs first, so a proof the verifier would reject costs nothing

A finding worth giving back

Three proofs captured at 13:04 UTC reverted when submitted at 18:38 UTC the same day.

Same receipt. Same Merkle root and siblings. Different continuity roots, because attestations had rolled into new 100-block checkpoints. Eight roots at capture, fifty-eight five hours later, for the same transaction.

What it cost to learn

210,630, 235,270 and 324,870 gas burned in the verifier before rejection. Re-fetched seconds before sending, the same three transactions recorded fine.

What nomen does about it

Every proof is fetched immediately before its send, gas is estimated first, and the query id is computed off-chain so anything already recorded is skipped. Zero reverts since.

The score

```
base          500
+ min(200, repays x 10)
+ 50  if borrows > 0 and liquidations == 0
- 150 per liquidation, decaying to 0 over
      ~1 year of Ethereum blocks since
      the last one
clamp 0..1000

grade  A >= 700   B >= 600   C >= 500
       D >= 350   otherwise E
```

It is not a credit model

It is a demonstration that one can be computed on-chain from inputs nobody had to be trusted about.

The formula is in the contract and in the README. The inputs are proven Ethereum transactions. A lender who disagrees with the weights can write their own function over the same records.

The decay is measured against the attested Ethereum head read from the ChainInfo precompile, not against wall-clock time.

What is not claimed

Compound v3 is excluded

Comet emits Supply and Withdraw; whether a withdraw is a borrow depends on the sign of the account's base balance, which is state. Attestcoin proves transaction inclusion, not state. Rather than guess, Compound is out of v1.

USD is an off-chain figure

Stablecoins at 1.00, WETH and wstETH and the BTC assets from Chainlink at one stated block. Everything else is counted in units only, and the page says how many events are priced.

The score is not a model

Five terms, stated in full. It exists to show the inputs can carry one.

Amounts in the contract are raw token units. Anything unmeasured says unmeasured.

Anyone can prove anything into it

Paste an Ethereum address on the site. It scans that address across the three protocols, fetches a proof for every event it finds, and records them. Your rows jump the queue.

Measured: an address scan over 50,000 blocks took 14 seconds and queued 916 new events. Self-submitted rows take at most half of each batch so one busy address cannot starve the market feed; liquidations lead regardless of source.

No permission

record is public. There is no allowlist, no owner and no key that can edit or delete what it wrote.

No forgery

A record you cannot produce a valid Ethereum proof for cannot be written, by us or by anyone.

Credit history the borrower owns

More protocols and more chains as Attestcoin adds them. A lender-side read contract so a Creditcoin lending market can price a loan from a proven record instead of a submitted one. Compound when a state proof exists.



Live `nomen.ochinimus.app`

Contract `0x0c019ee3298111b3b067ae1ba93f73d93ee5230e` on Creditcoin CC3 testnet

Code `github.com/seekdaseek/nomen`

Tests 24 Foundry tests, five against real prover responses

Author `ochinimus` · Chisinau · `ochinimus.app`